

MUSTAFA MUNIR

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EDUCATION

NED University of Engineering and Technology

October 2020 – July 2024

BS. Computer Science

Karachi

Completed a 4-year bachelor's degree in Computer Science & Information Technology from NEDUET, understanding and exploring the emerging technologies and methodologies for software development and computer architecture.

CGPA: 3.67/4.00

Salient courses: OOP, Web engineering, DBMS, DSA, PDC, Software Design.

DJ Sindh Govt. Science College

August 2018 – August 2020

Intermediate. Pre Engineering

Karachi

Pre-engineering from Karachi Board with **81.76%**.

WORK EXPERIENCE

SECURITI

Karachi

Associate Software Engineer

January 2025 – Present

Working as a Software Engineer focused on cloud data security and posture management utilizing Go, Python, and TypeScript.

- Implemented workerpool in Go to reuse 10 goroutines to process tasks, preventing excessive resource consumption.
- Developed 20+ cloud **metadata collectors** for posture data collection of GCP and AWS services.
- Wrote 2 post release **data migrations** for AWS ElastiCache and Neptune Graph Database to handle **data integrity issues**.
- Implemented 140+ compliance rules for a wide range of GCP and AWS cloud services.
- Triage and **fixed 25+ bugs** reported on **Production** and **QA** Environments.

Skills: Go, Python, TS, Postgres, Docker, Neo4j, GCP, AWS (Elasticsearch, Neptune, IAM), Redis, SQL, Design Patterns.

TECHWARDS

Karachi

Software Engineer

October 2023 – December 2024

Worked for 1 year and 4 months as a Software Engineer at Techwards on Generative AI driven backend systems.

- Engineered a Generative AI video creation pipeline integrating OpenAI for scriptwriting and Dall-E for image generation cutting video production cycles from 2 minutes to just 25 seconds.
- Implemented **automated workflows** for WebAffinity projects, integrating 2 webhooks that facilitated **video generation**, resulting in a 40% reduction in video production time.
- Played an important role with a team of 11 in a **blog-content generation** and updation tool for a publishing firm using **generative AI** with competitor analysis.
- Made 3 open source contributions using Java to project **Kestra** that we were using internally.
- Added test cases using Pytest & Swagger API docs to 2 projects.
- Done AML analysis of crypto currencies in Databricks for 2 clients.

Skills: Generative AI, Python, Django, Flask, SQLAlchemy, SMTP, DRF, Pytest, Langchain, Postgres, Vector Databases, Swagger, Postman, Docker, Black & Flake8, Github Workflows and CI/CD.

CONCEPT-ZI (CONTRACT)

Karachi

Software Engineer

May 2023 – September 2023

- As a backend engineer, completed 2 projects of domains: genetics engineering and space insights.
- Setup **Stripe payment gateway** and event webhooks for card based Visa and Mastercard transactions.
- Wrote automation script for data records migration from google sheets to **normalised database schema** containing 10+ tables.
- Converted nested **ORM** calls to raw query utilizing SQL clauses which **reduces query processing time** to 60%

Skills: Django, Python, ORM, Postgres, Git, AWS (S3, SES, Fargate), Selenium.

SASTATICKET

Karachi

Backend Engineering - Intern

January 2023 – March 2023

- Worked as Django Backend Intern.
- Developed Mock Bus Provider to be used by QA team.
- Added long polled functionality to 2 large data APIs.

Skills: Backend Development basics, API Testing, Postman Automation, API & Workflow Documentation.

PROJECTS

[Youtube Study Companion](#)

An open-source YouTube companion bot that lets users query large YouTube playlists and videos in natural language.

[SnagIt](#)

SnagIT is an AI-powered tool designed to automate the processing and triage of security alerts (offences) received from SIEM systems.

[Forward Proxy](#)

A basic HTTP proxy server with built-in features for request handling and response caching. It includes security features such as a forbidden host firewall and banned words detection.

[The Bronx Board](#)

An awareness-centric, AI-powered LMS dedicated to empowering Bronx youth to overcome societal challenges..

CERTIFICATIONS

[DATA ENGINEER ASSOCIATE](#)

(Proctored & Accredited)

Organization: **Databricks** Scored: **100%**

[CERTIFIED AGILE & SCRUM MASTER](#)

(Accredited)

Organization: **HRD University** Scored: **95%**